

The Equilibrium Effects of Vacancy Referrals

Gerard J. van den Berg*

Hanno Foerster[†]

February 12, 2026

Abstract

Job vacancy referrals (VRs) are used by public employment services to connect unemployed workers and firms that otherwise would not have matched. Existing quasi-experimental evidence shows that VRs increase the job finding probabilities of VR recipients. The impact of VRs on the economywide employment rate, however, depends on the magnitude of equilibrium effects. This paper studies the equilibrium effects of VRs and the channels through which they operate. We develop a search and matching model that incorporates several sources of equilibrium effects: congestion in the hiring process, disincentive effects on workers' job search effort, and changes in firms' vacancy posting. We estimate our model using German data on unemployed workers' job search behavior and firms' hiring decisions. Based on our estimated model, the individual-level effect of assigning a VR is 3.8 p.p., while the corresponding effect of a large-scale reform is about 10 percent smaller due to equilibrium spillovers.

Keywords: unemployment, vacancy referrals, wage, unemployment insurance, monitoring, moral hazard, structural estimation, counterfactual policy evaluation

JEL classification: J64, J65, C51, C54

*University of Groningen, University Medical Center Groningen, IFAU Uppsala, IAB Nuremberg, ZEW, CEPR.

[†]Boston College, IZA

We thank Axel Gottfries, Nicole Gürtzgen, Andreas Gulyas, Leo Kaas, Gesine Stephan, Michèle Tertilt and conference participants at the IAB-workshop “Vacancies, Hiring and Matching” for helpful comments. We are grateful to the IAB Institute of the German Federal Employment Agency, and especially to Nicole Gürtzgen and Gesine Stephan, for facilitating the project.

1 Introduction

In most OECD countries, public employment services (PES) provide job search assistance to unemployment benefit recipients. A common component of job search assistance is the use of vacancy referrals (VRs), which are directives to apply for specific job openings. PES use VRs both to support benefit recipients' own job search efforts and to encourage individuals whose search effort is deemed insufficient to apply for additional jobs (see, e.g., Grubb, 2001; Immervoll and Knotz, 2018). The overarching motivation for using VRs as part of active labor market policy (ALMP) is to bring together unemployed workers and firms that otherwise would not have matched.

An open question in evaluating VRs is the extent to which their impact is attenuated by equilibrium effects.¹ In this paper, we study three channels through which equilibrium effects may arise. First, we examine whether assigning VRs to a group of unemployed workers reduces the job-finding prospects of other unemployed workers. Second, we analyze how VRs affect unemployed workers' own job search effort. Third, we study how firms' vacancy posting behavior responds to changes in the use of VRs.

Existing empirical evidence suggests that VRs have a positive impact on the job-finding probabilities of VR recipients (see, e.g., Bollens and Cockx, 2017; van den Berg, Hofmann, and Uhlendorff, 2014).² Further evidence is provided by Fougère, Pradel, and Roger (2009), who estimate a partial equilibrium job search model to study the extent to which VRs disincentivize job search effort. They find moderate disincentive effects, but show that the net employment effect of VRs remains positive and large in magnitude.

Our goal in this paper is to investigate the extent to which the positive effects of VRs on individual job finding probabilities translate into an increase in the economywide employment rate. To this end, we develop and estimate a dynamic model that accounts for several channels through which equilibrium effects may operate.

Our model reflects the fact that unemployed workers submit multiple job applications, while typically searching for only one job at a time. Similarly, firms face a queue of multiple applications for a given vacancy. When a worker receives a VR and applies, an additional application is added to the queue for that vacancy, which typically reduces the hiring probability of all other

¹See Abbring and Heckman (2007) for a survey of the literature on general equilibrium evaluation of ALMPs.

²Using data from Flanders, Bollens and Cockx (2017) find that VRs substantially increase job-finding probabilities in both the short and the long run. Using German data, van den Berg, Hofmann, and Uhlendorff (2014) show that VRs are associated with higher job-finding probabilities. Using data from Sweden, Engström, Hesselius, and Holmlund (2012) find that only about one third of VRs result in job applications, but that intensified monitoring increases the share of VRs that lead to applications.

applicants. We refer to this mechanism as congestion in the hiring process. Such congestion may result in crowding out, whereby a referred applicant displaces another applicant who would otherwise have been hired. As a result, the net effect of VRs on the economywide employment rate may be zero if a VR simply crowds out another applicant who, as a consequence, remains unmatched.

At the other extreme, VRs may lead to the filling of vacancies that would otherwise have remained unfilled, either because no other workers applied or because displaced applicants find employment elsewhere. Our structural model formalizes this reasoning and allows us to quantify the extent of congestion and the resulting crowding-out effects. In addition, the model accounts for endogenous changes in firms' vacancy posting behavior in response to VRs, as well as for potential disincentive effects of VRs on workers' own job search effort, as in Fougère, Pradel, and Roger (2009).

More specifically, we develop a dynamic search and matching model in which unemployed workers and firms match at random according to an urn-ball matching technology (see, e.g., Rogerson, Shimer, and Wright (2005) and Petrongolo and Pissarides (2001)). Workers send out multiple applications to a (random) set of firms, and each firm hires at most one worker from its applicant pool.³ Firms optimally choose whether to post a vacancy and which wage to post. They take the resulting applicant pool as given and select one worker from it when filling the vacancy.

In each period, unemployed workers choose how many applications to send to firms and may additionally receive a VR from the PES. Receiving a VR increases the job-finding probability of the worker who receives it, but it also increases congestion in the applicant pool of the targeted vacancy, thereby reducing the hiring probabilities of other applicants. The resulting degree of crowding out depends on labor market tightness and on the number of applications sent by the VR recipient and by other workers applying to the same vacancy. In addition to congestion effects, changes in the number of VRs in the economy affect equilibrium outcomes by influencing firms' vacancy and wage posting decisions as well as workers' incentives to submit applications.

We estimate the model using the generalized method of moments (GMM), targeting moments from German survey microdata on unemployed workers' search behavior and post-unemployment outcomes, as well as from data on firms' hiring decisions. The worker-side data are informative, among other things, about the number of applications sent and VRs received for an inflow sample into unemployment in 2008. The firm-side data contain information on the number of hires and

³Our model builds on the search and matching framework with multiple applications developed in Albrecht et al. (2004), Albrecht, Gautier, and Vroman (2006), and Gautier, Moraga-González, and Wolthoff (2016).

unsuccessful searches in the previous year, as well as detailed information on the applicant pool for the firm’s most recent job opening, including the number of candidates referred through a VR and whether the hired candidate was referred through a VR.

We use the estimated model to conduct counterfactual policy experiments in which the VR rate (the share of workers receiving a VR in a given period) is varied. Based on these policy simulations, we study the impact of VRs on the economywide employment rate and quantify the magnitude of equilibrium effects. In particular, we use the model to assess the extent to which individual-level treatment effects that ignore equilibrium effects would overstate the impact of large-scale policy reforms that increase the VR rate on worker match rates.

We find that equilibrium effects substantially reduce the effectiveness of VRs. Under the status quo policy, the VR rate is 56%, meaning that 56% of unemployed workers receive a VR in a given month. Based on our estimated model, a large-scale reform that increases the VR rate from 56% to 90% would raise the worker match rate by 0.9 percentage points. A decomposition of this effect shows that, absent equilibrium effects, the impact of the reform on the worker match rate would be considerably larger, at 1.3 percentage points, implying that equilibrium effects reduce the impact of the reform by about 30%.

Relatedly, the model-implied treatment effect of assigning a VR to a single individual equals 3.8 percentage points. In contrast, the average effect of the large-scale reform on newly treated individuals is about 10% smaller, at 3.4 percentage points. Accordingly, estimates from small-scale field experiments that ignore equilibrium effects would overstate the effect of such a reform by about 10%. Moreover, the reform reduces the match rates of individuals who were already treated prior to the reform as well as those of untreated individuals.

The contribution of this paper is twofold. First, we develop a structural model of the labor market that incorporates VRs and accounts for three main channels through which equilibrium effects may operate: congestion effects in the hiring process, disincentivized job search effort, and endogenous changes in firms’ vacancy posting behavior. Second, we estimate the model using worker- and firm-level data and use the estimated model to quantify the discrepancy between the individual-level effects of VRs documented in previous empirical studies and the aggregate effect of VRs on the economywide employment rate.

We thereby contribute to a small literature that studies the equilibrium effects of ALMPs. In particular, Crépon et al. (2013) and Gautier et al. (2018) provide empirical evidence on the equilibrium effects of ALMPs.⁴ Crépon et al. (2013) interpret their empirical findings through the lens of a stylized search and matching model in which equilibrium effects arise exclusively

⁴In these papers, the ALMPs considered include job search assistance, training programs, as well as VRs.

from firms' vacancy posting responses to changes in ALMPs. Gautier et al. (2018) estimate a structural model to analyze counterfactual policy scenarios in which search costs are reduced. Relative to these papers, we provide the first study of the equilibrium effects of VRs and the channels through which these equilibrium effects operate.

The congestion effects we study are related to congestion on online dating platforms, where users send many essentially costless signals (Lehmann, Terrier, and Lalive, 2023). A key difference in labor markets is that only workers send applications, whereas firms post vacancies and select applicants.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 develops the model, Section 4 describes the data, and Section 5 outlines the model estimation. Section 6 presents the results from the policy simulations. Section 7 concludes.

2 Institutional Background

In most OECD countries recipients of unemployment insurance benefits are subject to a range of job search monitoring and job search assistance practices (see Grubb, 2001). VRs are one of the most important of these practices (see Immervoll and Knotz, 2018). By making use of VRs unemployment insurance agencies seek to support individual benefit recipients' own job search effort, but also to activate benefit recipients whose own search effort is deemed insufficient.

In Germany UI benefit recipients may receive VRs starting immediately after they start claiming benefits. Caseworkers at the public employment services (*Bundesagentur für Arbeit*) select job vacancies from the agency's database and matches them to UI benefit recipients.⁵ The selected benefit recipients are then notified by regular mail about the vacancy, are typically given a brief description about the job opening and are asked to apply for the referred position(s) (this description typically does not include the wage).⁶ Not applying for a referred vacancy or rejecting a resulting job offer may lead to sanction (a temporary benefit reduction), although enforcement of such sanctions is infrequent (see van den Berg, Foerster, and Uhlendorff, 2019).

There are two standard procedures by which job openings may enter as vacancies into the database of the PES. Firms may directly contact the PES and request that a specific vacancy is used for making VRs. Moreover, to find suitable vacancies caseworkers browse public job advertisements such as newspaper- and online-ads, i.e., in principal most publicly advertised job

⁵The same vacancy may be referred to several UI benefit recipients and benefit recipients sometimes receive several VRs at the same time.

⁶A VR typically contains information about the occupation, the working hours and the date of potential job start (see van den Berg, Hofmann, and Uhlendorff, 2014).

openings may end up being used by the PES for making VRs. There are two restrictions that caseworkers need to abide when making a VR. The job has to be within 2.5 hours commuting distance and has to pay at least the same amount that the benefit recipient receives in UI benefits. There are no other restrictions on the type of jobs that a UI benefit recipient may receive as VRs. In particular job openings referred as VRs do not need to match the pre-unemployment occupation of the UI benefit recipient who is assigned the vacancy.

3 Model

Our model is a dynamic search and matching model in which unemployed workers in each time period send out multiple applications, which are received by firms. Each time period each firm makes one job offer to one worker in their applicant pool. They decide which wage to offer to this worker and if the worker accepts a match is created and the worker starts producing at this firm. If the worker rejects the wage offer, either because he received a preferable offer from another firm or because staying unemployed is preferable to him, the firm's position remains vacant for one time period. VRs are applications made by the unemployment agency on behalf of individual workers. There are several equilibrium effects and margins of crowding out that our model accounts for. First, A VR *ceteris paribus* increases the probability of being matched for the worker who receives the VR, but incurs a negative externality by decreasing the likelihood of being matched for all other unemployed workers. Second, VRs may crowd out individual job search effort, i.e., upon receiving a VR an unemployed worker may decide to write less job applications. Third, there is free entry and firm entry and wages may respond to the number of applications and VRs in the economy.

Formally, our model is set in discrete time and the time horizon is infinite. There is a unit mass of workers in the economy and a measure v of vacancies, which is endogenously determined. All agents have linear preferences and discount the future at rate β . Firms are homogeneous and workers are heterogeneous in their costs of writing applications, but are homogeneous in their productivity, i.e., whenever a firm and a worker are matched they produce a constant per period output y .

Hiring process In each time period firms receive multiple applications from different workers. We assume that search is undirected, i.e. each firm is equally likely to receive a given application. Denote by $p_{\alpha,\iota}$ the population share of workers writing $\alpha \in \{0, 1, 2, \dots, A\}$ applications and receiving $\iota \in \{0, 1, 2, \dots, J\}$ VRs. The queue length, i.e., the expected total number of applications

for a given firm is⁷

$$\begin{aligned}\lambda &= \frac{u}{v} \sum_{\iota=0}^J \sum_{\alpha=0}^A (\alpha + \iota) p_{\alpha, \iota} \\ &= \frac{u}{v} \sum_{\iota=0}^J p_{\iota} \sum_{\alpha=0}^A (\alpha + \iota) p_{\alpha | \iota}.\end{aligned}\tag{1}$$

As workers are homogeneous in their productivity, firms pick a candidate at random. **AlbrechtEtal04** show that in this case the probability that a given application is successful is⁸

$$\psi = \frac{1}{\lambda} (1 - e^{-\lambda}).$$

For a worker who sends out α applications and receives ι VRs the the probability to receive $k < \alpha + \iota$ job offers then is given by

$$\mu(k | \alpha + \iota) = \binom{\alpha + \iota}{k} \psi^k (1 - \psi)^{\alpha + \iota - k}.$$

Workers When unemployed, workers receive unemployment benefits b per period. Each period they choose how many applications to write and upon receiving job offers, whether to accept or reject. The probability of receiving k job offers, $\mu(k | \alpha + \iota)$, depends on the sum of number of job applications that an unemployed worker sends out, α , and the number of VRs he receives, ι .

Upon writing α job applications and receiving ι VRs the unemployed worker incurs application costs $C(a, \iota) = c \cdot (\alpha + \phi \iota)^2$. c , a parameter that scales the application costs, is heterogeneous in the worker population and is redrawn in each time period (i.e. for a given worker c is iid across time). We assume $0 < \phi < 1$, i.e., applying for VRs is costly, but less costly than applying for jobs that are not referred by the public employment services. The underlying assumption is that applying for VRs is costly to some extent, as time costs and psychic costs of preparing for a job interview are incurred. We argue however that applying for VRs is less costly than applying for non-referred jobs, as the search process of looking through job ads, figuring out which jobs are suitable and deciding where to apply is undertaken by the public employment services.

The Bellman equation for an unemployed worker conditional on the number of applications

⁷The total number of applications is the sum of regular applications and VRs that a firm receives.

⁸For a detailed exposition of urn-ball matching with multiple applications see Albrecht, Gautier, and Vroman (2006).

and VRs, α and ι , and the application cost parameter c is

$$V_U(\alpha|c, \iota) = b - C(\alpha, \iota) + \beta \left(\sum_{k=1}^a \mu(k|\alpha, \iota) \int_0^\infty \max\{V_U, V_E(w)\} dF^k(w) + \mu(0|\alpha, \iota) V_U \right) \quad (2)$$

where $F(w)$ is a wage offer distribution that workers take as given. For a given worker the optimal number of applications is

$$\alpha(\iota, c) = \operatorname{argmax} V(\alpha|c). \quad (3)$$

The unconditional value of unemployment is given by the Bellman Equation that takes into account that workers optimally choose the number of applications they send out and computed prior to the realization of the search cost parameter, i.e., taking expectations over c .

$$V_U = \int_0^\infty \sum_{\iota=0}^J p_\iota \max_{\alpha} V(\alpha|c, \iota) dS(c) \quad (4)$$

In terms of timing we assume that in a given time period workers decide how many applications to send out after learning the number of VRs they receive.

Employed workers receive a constant wage w each time period for the duration of their employment spell and are layed off at rate δ . The value of employment thus is given by

$$V_E(w) = w + \beta((1 - \delta)V_E(w) + \delta V_U) \quad (5)$$

It is straightforward to show that V_E is strictly increasing in w . A workers decision rule in accepting and rejecting job offers thus can be summerized by a reservation wage $\bar{w}$.

Using (5) together with (4) and rearranging, yields the reservation wage equation

$$\bar{w} = b + \beta \int_0^\infty \max_{\alpha} \left(\sum_k^{A+J} \mu(k|\alpha + \iota) \int_{\bar{w}}^\infty \frac{w - \bar{w}}{1 - \beta(1 - \delta)} dF^k(w) - C(\alpha, \iota) \right) \quad (6)$$

Firms Firms in each time period decide whether to post a vacancy and, if they decide to post a vacancy, which wage to post. The value a firm attributes to being matched with a worker and paying wage w is

$$\begin{aligned} V_M(w) &= y - w + \beta(1 - \delta)V_M(w) \\ &= \frac{y - w}{1 - \beta(1 - \delta)} \end{aligned} \quad (7)$$

where y is the productive value of a match. The value of posting a vacancy is given by

$$V_F(w) = -k + \beta m_F(w) V_M(w),$$

where k is the cost of posting a vacancy and $m_F(w)$ is the endogenous probability for a given firm to be matched with a worker. Gautier, Moraga-González, and Wolthoff (2016) show that the matching probability for a firm is given by

$$m_F(w) = \frac{u}{v} \sum_{k=1}^{A+J} k q_k F^{k-1}(w),$$

where q_k is the population share of unemployed workers receiving k job offers,

$$\begin{aligned} q_k &= \sum_{\iota=0}^J \sum_{\alpha=0}^A \mu(k|\alpha, \iota) p_{\alpha, \iota} \\ &= \sum_{\iota=0}^J p_{\iota} \sum_{\alpha=0}^A \mu(k|\alpha, \iota) p_{\alpha|\iota}. \end{aligned} \tag{8}$$

We assume free market entry, i.e. it is optimal for new firms to enter the market and post vacancies until the value of posting a vacancy is driven down to zero, i.e., until $V_F(w) = 0$ for all w or equivalently

$$\begin{aligned} 0 &= -k + \beta m_F(w) V_M(w) \\ &= -k + \beta m_F(w) \frac{y - w}{1 - \beta(1 - \delta)}, \end{aligned} \tag{9}$$

where the second equality is obtained by inserting (7). From a firms perspective it is never optimal to post wages below the workers reservation wage $\bar{w}$. Denote the smallest wage posted with positive probability by $w_{\min}$. This wage equals the reservation wage, i.e., $w_{\min} = \bar{w}$. The firm posting the smallest wage will only attract unemployed workers who receive no other job offer, i.e., $m_F(\bar{w}) = q_1$. Using $V_F(w) = V_F(\bar{w}) = 0$ can be used together with $m_F(\bar{w}) = q_1$ to characterize the wage offer distribution by

$$\sum_{k=1}^{A+J} k q_k F^{k-1}(w) (y - w) = q_1 (y - \bar{w}). \tag{10}$$

For given y , q_k and $\bar{w}$, (10) can be used to solve for the wage offer distribution F pointwise at any w .

Steady State For the economy to be in steady state, flows into unemployment need to equal flows out of unemployment, i.e.,

$$0 = \delta e - m_W u, \quad (11)$$

so that the steady state unemployment is given by $u = \frac{\delta}{\delta + m_W}$.

Equilibrium A steady state equilibrium of the economy is given by a queue length λ , a policy function, $\alpha(\iota, c)$, a reservation wage $\bar{w}$, a measure of vacancies v and a wage offer distribution F , such that (1), (3), (6), (9), (10) and (11) are satisfied.

Distribution of Accepted Wages In the described setup a closed form relationship between the equilibrium wage offer distribution F and the distribution of accepted wages can be derived. Making use of this relationship will allow us to compute moments from the accepted wage distribution analytically. This will be useful for the structural estimation of the model as it allows us to target moments of the distribution of accepted wages rather than moments of the (unobserved) wage offer distribution. Denote the distribution of accepted wages by $G(w)$. Then in equilibrium it holds that

$$G(w) = \frac{\sum_{k=1}^{A+J} q_k F^k(w)}{1 - q_0}$$

(see Gautier, Moraga-González, and Wolthoff, 2016). We use this relationship in computing moments of the distribution of accepted wages implied by our model by using $F(w)$ to compute $G(w)$ at a set of approximation points and applying numerical integration techniques to obtain the desired moments.

4 Data

We estimate our model using a combination of worker and firm data. For worker-side information we use the IZA Evaluation Dataset, a survey covering a large sample of workers who became unemployed between June 2007 and May 2008 in Germany.⁹ The survey covers a random sample of 17,396 newly unemployed workers aged 16 to 54, who are surveyed up to three times: once as soon as possible after inflow into unemployment, again roughly one year later, and in a final wave approximately three years after the first interview.

The data contain detailed information on individuals' labor market histories and job search

⁹See Caliendo et al. (2011) and Arni et al. (2014) for a detailed description of the IZA Evaluation Dataset.

behavior, including the number of applications submitted, vacancy referrals received, and monthly unemployment benefits during the last unemployment spell before the survey interview, as well as the start and end dates of all unemployment and employment spells since inflow. In addition, the data include basic sociodemographic characteristics such as age, gender, region of residence, and highest educational attainment.

For information on firm hiring we use the IAB Job Vacancy Survey, a representative survey among German establishments that collects detailed information on recruiting processes.¹⁰ The survey is conducted as a repeated cross section and covers approximately 10,000 to 16,000 firms per year between 2000 and 2014.

The data contain detailed information on firms' hiring and recruitment activities, including the number of applications and vacancy referrals received for the most recently filled vacancy, the number of hires and unsuccessful search attempts over the past year, the minimum educational degree required for applicants for the last filled vacancy, as well as the firm's industry sector and region of location.

Sample In selecting our sample we make use of a narrow time window in which our two data sources overlap. The IZA Evaluation Dataset covers workers who became unemployed in 2007 and 2008 and follows them for up to three years. The IAB Job Vacancy Survey covers the period from 2000 to 2014. However, a question on how often firms were unsuccessful in filling a vacancy, a key variable in our estimation, was introduced only in 2008. For the structural estimation of our model, we therefore restrict our attention to 2008, the only year for which all relevant information from both datasets is available.¹¹

For the IZA Evaluation Dataset, we begin with a sample of 6,520 workers who entered unemployment in 2008. We restrict the sample by excluding individuals younger than 18 and observations with reported unemployment benefits, numbers of monthly applications, or numbers of monthly vacancy referrals above the respective 98th percentiles. We further exclude workers with reported monthly income above the 98th percentile or below the 2nd percentile. The resulting analysis sample consists of 6,174 individuals and 17,219 person-year observations.

Wages and industry sector are observed for individuals who experience at least one employ-

¹⁰See Kubis (2018) for a detailed description of the IAB Job Vacancy Survey. The unit of observation in the IAB Job Vacancy Survey is the establishment. For simplicity, we refer to establishments as firms throughout the paper.

¹¹It is worth noting that although our sample period borders on the onset of the global financial crisis, the German labor market proved surprisingly robust during the 2009 recession, with unemployment and wages remaining relatively stable (Burda and Hunt, 2011; Möller, 2010). This resilience has often been attributed to wage moderation in the years preceding the recession as well as to the extensive use of short-time work arrangements (Boeri and Bruecker, 2014).

ment spell between inflow and the third survey wave. This applies to 4,118 individuals in our analysis sample (66%).

Our initial sample from the IAB Job Vacancy Survey consists of 13,652 firms surveyed in 2008. We exclude firms reporting numbers of applicants or hires above the respective 98th percentiles. The resulting analysis sample covers 13,428 firms.

Descriptives To provide an overview of the German labor market in 2008, we report summary statistics on the search behavior of unemployed workers in Table 1 and on the hiring behavior of firms in Table 2.

Table 1 reports descriptive statistics for the IZA Evaluation Dataset sample. The summary statistics indicate that the average accepted wage of individuals who exit unemployment substantially exceeds the average monthly unemployment benefits claimed by unemployed individuals, by a factor greater than 1.7, suggesting a large employment premium. Unemployed individuals in our sample send an average of 12.4 applications and receive 1.9 vacancy referrals (VRs) in the month prior to the survey. Interestingly, individuals who receive at least one VR report, on average, 2.8 more applications than those who do not receive a VR, suggesting that crowding out of job search effort by VRs is not a dominant force.¹² Finally, individuals in our sample are on average 36 years old, and 48% are female.

Table 2 provides summary statistics on firms' hiring decisions from the 2008 wave of the IAB Job vacancy survey. The displayed descriptive statistics show that few candidate searches are unsuccessful and need to be cancelled without filling a vacancy, relative to the number of searches that end in a hire. It is implied by these statistics that around 31% of all candidate searches are unsuccessful. On average firms report that for the last vacancy filled before the survey interview 15.5 applications were received 4.9 of which were referred by the PES as VRs. Thus on average 31.8% of all received applications were received through VRs.

¹²The positive correlation between the number of applications and VRs, $\text{Cov}(\alpha, \iota) = 0.1$, points in the same direction.

Table 1: Summary statistics, IZA Evaluation Dataset

Variable	Mean	Std. Dev.
Unemployment benefits	634	323
Accepted wage	1111	549
# applications	12.43	12.40
# applications (if VRs=0)	10.82	11.79
# applications (if VRs>0)	13.67	12.71
# VRs	1.94	2.78
Age	35.54	10.56
Female	0.48	0.50

Notes: Summary statistics based on the 6,174 individuals in our IZA Evaluation Dataset analysis sample. Statistics on *Age* and *Female* are averages over the full sample. Statistics on *Accepted wage* are conditional on employment and statistics on *Unemployment benefits*, *# Applications* and *# VRs* are conditional on unemployment. Data on *accepted wages* correspond to monthly wages in euros. Data on *# Applications* and *# VRs* correspond to applications sent and VRs received in the month prior to the survey interview.

Table 2: Summary statistics, IAB Job Vacancy Survey

Variable	Mean	Std. Dev.
# unsuccessful searches	1.72	0.02
# hires	3.86	0.09
# applications received (last vacancy)	15.52	0.64
# VRs received (last vacancy)	4.93	0.31

Notes: Summary statistics from the 2008 wave of the IAB Job Vacancy Survey. The no. of hires and unsuccessful searches relate to the last 12 months before the survey interview. Data on *applications received* and *VRs received* relate to the last vacancy filled at the surveyed firms. The number of applications received includes both VR and non-VR applications.

Figure 1 illustrates the distributions of the number of monthly applications sent and VRs received by workers in panels A and B, and the distributions of applications and VRs received by firms for the last vacancy they filled in panels C and D.

Panel A shows substantial bunching at round numbers in workers' reported numbers of ap-

plications, particularly for values above 10. In our sample, 6% of workers report having sent no applications. The median of the distribution is 8 applications and a small share of workers (less than 5%) report having sent out more than 40 applications.

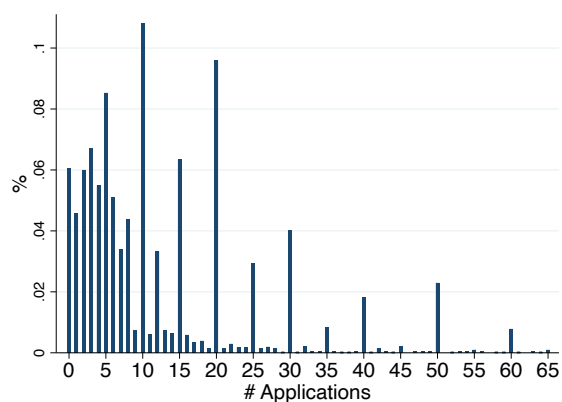
Panel B shows that 44% of unemployed workers report not having received a VR in the month prior to the interview, while 17% report having received exactly one. The 95th percentile of the distribution is 8 VRs.

Panel C shows that 19% of firms report having received only one application for their most recent job opening. The median and the 90th percentile of the distribution are 4 and 26 applications, respectively. The distribution exhibits strong right skewness. 3.5% of firms reporting more than 40 applications for their most recent job opening.

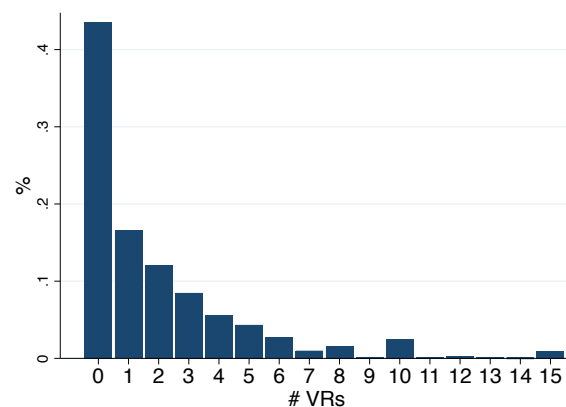
Panel D shows that a large share of firms (31%) report having received no VR-applications. Extremely few firms (less than one percent) report having received more than 15 VR-applications for their most recent job opening. 42% report having received 1-5, 15% report 6-10 and 6% report having received 10-15 VR-applications.

Figure 1: No. of Vacancy Referrals and Applications, Worker Side

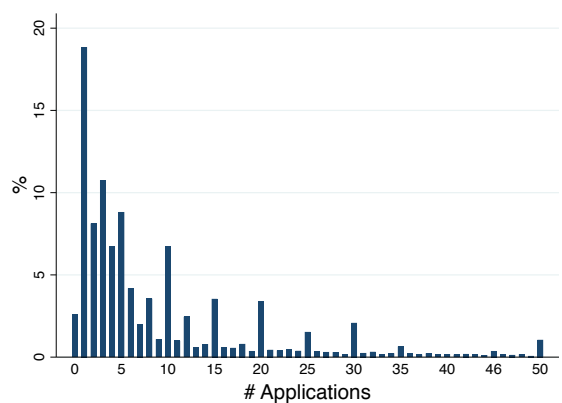
Panel A: No. of applications sent out per month (worker side)



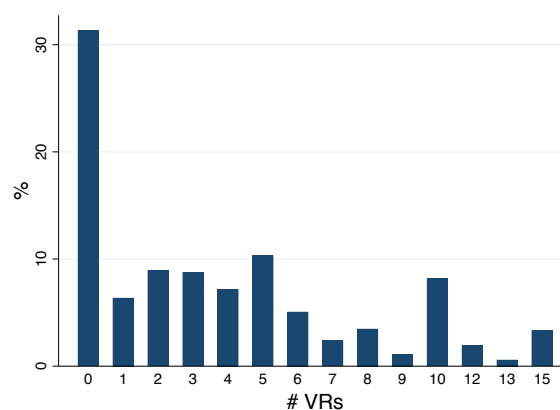
Panel B: No. of VRs received per month (worker side)



Panel C: No. of applications received per vacancy (firm side)



Panel D: No. of applicants via VRs per vacancy (firm side)



Notes: Panels A and B are based on our analysis sample of 6,174 individuals from the IZA Evaluation Dataset. Panels C and D are based on our analysis sample of 13,428 firms from the IAB Job Vacancy Survey.

5 Estimation

To obtain estimates of the structural parameters of the model, we proceed in three steps. First, a subset of parameters is set externally to match values from the existing literature and external data sources. Second, several parameters are fixed to match their empirical counterparts measured in the IZA Evaluation Dataset. Finally, the remaining parameters are estimated using the generalized method of moments (GMM). Specifically, we search over the parameter space to find values that minimize the distance between moments computed from the structural model and their empirical counterparts.

5.1 Pre-set and directly estimated parameters

The parameters that are set externally are summarized in Table 3. We fix the monthly discount factor at $\beta = 0.997$. Monthly unemployment benefits are set to $b = 634$ corresponding to the average monthly unemployment insurance benefits observed in our IZA Evaluation Dataset sample. The maximum number of monthly applications is set to $A = 60$, which corresponds to the 98th percentile of the empirical distribution of monthly applications in our IZA Evaluation Dataset sample. We furthermore set $J = 1$, imposing that workers receive at most one VR. Under this restriction, a policy scenario is fully characterized by the fraction of unemployed workers who receive a vacancy referral in a given month. A policy scenario in this setting thus is fully characterized by the fraction of unemployed workers who receive a VR in a given month, i.e., the VR rate. We set the VR rate equal to 0.56, the fraction of workers who report having received at least one VR last month in our IZA Evaluation Dataset sample.

Table 3: Externally set parameters

Parameter	Value
Monthly discount factor, β :	0.997
Monthly unemployment benefits, b :	634
Max. number of monthly applications, A :	60
Max. number of monthly VRs, J :	1
VR rate, $p_{vr} = P(\iota = 1)$	0.56

5.2 Generalized method of moments estimation

The remaining parameters, which we estimate using the generalized method of moments (GMM), are the vacancy creation cost, k , the match flow value y , the VR application cost scale parameter ϕ , and the job destruction rate δ . We further assume that application costs, c , are log normally distributed, $\ln(c) \sim \mathcal{N}(\mu, \sigma)$, and estimate μ and σ . We denote the vector of these parameters by $\theta = (k, y, \phi, \delta, \mu, \sigma)$. For a given parameter vector θ , we solve the structural model using dynamic programming.¹³ Having solved the model, we compute a vector of model-implied moments, $m(\theta)$. GMM-estimation minimizes the distance between the vector of model-implied moments, $m(\theta)$, and its data counterpart, $\hat{m}$, i.e.,

$$\hat{\theta} = \operatorname{argmin}_{\theta} (m(\theta) - \hat{m})' \widehat{W} (m(\theta) - \hat{m}), \quad (12)$$

where $\hat{\theta}$ denotes the estimator of the structural model parameters and $\widehat{W}$ is a diagonal weighting matrix with elements given by the inverse variances of the empirical moments $\hat{m}$.

In the estimation, we target 18 empirical moments that can be approximated or computed directly as functions of the structural parameters. These moments include the shares for which α lies in each of 12 disjoint intervals (12 moments), the mean number of applications sent by workers who received a VR ($\iota = 1$) and who did not ($\iota = 0$) (2 moments), and the mean of accepted wages (1 moment). We further target the average number of applications received by firms per vacancy, the employer match rate, and the unemployment rate (3 moments).

5.3 Parameter Estimates and Model Fit

The GMM parameter estimates are presented in Table 4. The estimated vacancy referral application cost scale parameter, $\hat{\phi} = 0.39$, implies that the application costs for jobs that are referred through a VR are 39% of the application costs for non referred jobs. We estimate the per period output of a successful match to be $\hat{y} = 1039$. The implied reservation wage at the estimated parameter values is $\bar{w}(\hat{\theta}) = 830$. The estimated job separation rate, $\hat{\delta} = 0.04$ implies that matches on average last around 25 months. The costs of creating a vacancy, which in equilibrium equals the value of posting a vacancy, is estimated at $\hat{k} = 1389$.

¹³To implement the model numerically, we approximate the wage offer distribution by a piecewise linear cumulative distribution function, discretize the application cost distribution, and use numerical integration to approximate integrals.

Table 4: GMM parameter estimates

Parameter	Estimate
ϕ	0.39
δ	0.04
k	1389
y	1039
μ	1.7
σ	2.06

To assess the model fit Table 5 and Figure 2 compare targeted empirical moments with their model counterparts evaluated at the estimated parameter values.

The estimated model provides a good fit for most of the moments reported in Table 5. However, the model underpredicts the standard deviation of wages and the employer match rate. Moreover, by construction the model predicts that unemployed workers who receive a VR submit fewer applications than those who do not, $\mathbb{E}[\alpha|\iota = 0] > \mathbb{E}[\alpha|\iota = 1]$, whereas the opposite relationship is observed in the data.

Figure 2 further shows that the model captures the distribution of applications sent by unemployed individuals remarkably well.

6 Policy Simulations

To study the impact of VRs on labor market outcomes and to quantify their equilibrium effects, we consider a set of counterfactual policy scenarios. In the model, a policy is characterized by the VR rate, defined as the probability that a worker receives a VR in a given period, $p_{vr} = P(\iota = 1)$.¹⁴

6.1 The impact of VRs on aggregate labor market outcomes

We solve the model for a range of values of p_{vr} spanning the unit interval, holding all other model parameters fixed. Figure 3 shows how changes in the VR rate affect a set of aggregate labor market outcomes.

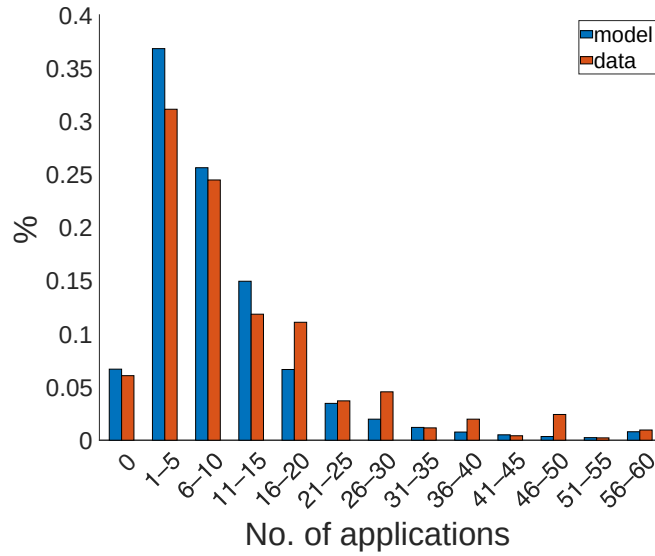
¹⁴Recall that we impose that a worker can receive at most one VR in a given time period ($J = 1$).

Table 5: Model fit 1

Moment	Data	Model
Queue length	10.59	11.49
Unemployment rate	7.8%	8.4%
$\mathbb{E}[\alpha \iota = 0]$	10.8	9.5
$\mathbb{E}[\alpha \iota = 1]$	13.7	9.3
Avg. accepted wage	1111	918
S.d., accepted wages	549	37
Employer matching rate	0.69	0.56

Notes: Displayed are moments implied by our model at the estimated parameter values and their empirical counterparts computed from the IZA Evaluation Dataset and the IAB Job Vacancy Survey Data.

Figure 2: Model fit 2, no. of applications



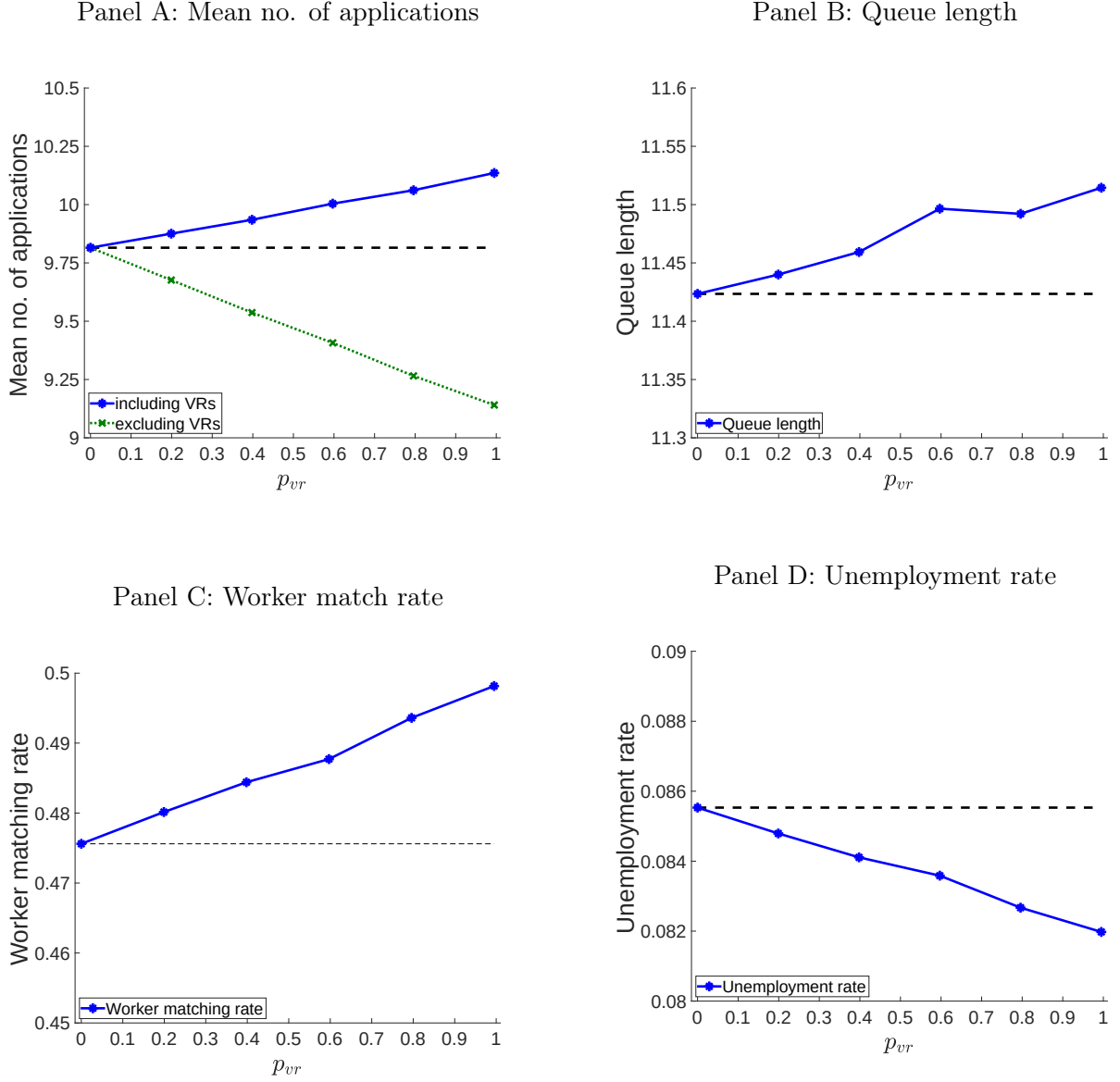
Notes: The figure displays the relative frequencies of workers by number of applications, as implied by our model evaluated at the estimated parameter values and their empirical counterparts computed from our IZA Evaluation Dataset analysis sample.

Panel A shows that increasing the vacancy referral rate has a positive effect on the mean total number of applications (including VR applications) submitted by workers. Moving from a scenario in which no workers receives VRs ($p_{vr} = 0$), to one in which all workers receive a VR with certainty in each period ($p_{vr} = 1$), increases the mean number of applications (including VR applications) by 0.3. Panel A also reports the mean number of applications not brokered through vacancy referrals. This measure declines with the vacancy referral rate, indicating that unemployed workers reduce the number of non referral applications they submit when they receive a vacancy referral. This behavioral response partially offsets the the positive effect of VRs on the overall number of applications, implying that the increase in total applications associated with a policy change from $p_{vr} = 0$ to $p_{vr} = 1$ is smaller than one.

Panel B shows that queue length increases with the vacancy referral rate, i.e., higher vacancy referral rates are associated with greater competition for a given vacancy faced by each applicant.

Panel C and D display the average worker match rate and the aggregate unemployment rate for different values of p_{vr} . Note that the worker match rate and the aggregate unemployment rate are linked through a nonlinear relationship given by equation (11). Panel C and D show that increasing the VR rate lowers the aggregate unemployment rate and increases the worker match rate, although the effect on the aggregate unemployment rate is quantitatively modest: Increasing the VR rate from $p_{vr} = 0$ to $p_{vr} = 1$ leads to a drop in the unemployment rate from 8.6% to 8.2%.

Figure 3: Counterfactual policy scenarios



6.2 Quantifying the equilibrium effects of VRs

To quantify the equilibrium effects of VRs, we decompose the aggregate effect of an increase in the VR rate on the worker match rate. To this end, the aggregate worker match rate, $m_W(p_{vr})$, can be written as a weighted average of the match rate for workers who receive a VR, $m_{W|vr=1}(p_{vr})$, and for those who do not, $m_{W|vr=0}(p_{vr})$

$$m_W = p_{vr}m_{W|vr=1} + (1 - p_{vr})m_{W|vr=0}, \quad (13)$$

where both conditional match rates are endogenous to the VR rate due to general equilibrium effects. Using this representation, the effect of a policy reform changing the VR rate from p_{vr} to

p'_{vr} can be decomposed as

$$\begin{aligned}
m_W(p'_{vr}) - m_W(p_{vr}) &= (p'_{vr} - p_{vr}) \underbrace{\left[m_{W|vr=1}(p_{vr}) - m_{W|vr=0}(p_{vr}) \right]}_{\mathbf{A}} \\
&\quad + p'_{vr} \underbrace{\left[m_{W|vr=1}(p'_{vr}) - m_{W|vr=1}(p_{vr}) \right]}_{\mathbf{B}} \\
&\quad + (1 - p'_{vr}) \underbrace{\left[m_{W|vr=0}(p'_{vr}) - m_{W|vr=0}(p_{vr}) \right]}_{\mathbf{C}}. \tag{14}
\end{aligned}$$

The first component captures the direct treatment effect of assigning vacancy referrals to additional unemployed individuals, holding equilibrium conditions fixed. It reflects the difference in match rates between individuals who receive a vacancy referral and those who do not, evaluated at the pre-reform equilibrium. The second and third components capture equilibrium effects on unemployed individuals whose treatment status does not change but who are nevertheless affected by the policy through general equilibrium adjustments. Together, these components reflect equilibrium spillovers arising because changes in the vacancy referral rate affect application behavior, congestion in the hiring process, and firms' vacancy posting decisions, thereby altering matching probabilities even for workers who are treated both before and after the reform or untreated both before and after the reform.¹⁵

Figure 4 applies the decomposition to our estimated model and illustrates the magnitudes of the components. The figure reports the decomposition for a change from the status quo policy, $p_{vr} = 0.56$, to $p'_{vr} = 0.9$. We find that this increase in the VR rate raises the worker match rate by 0.9 percentage points overall, reflecting a positive treatment effect ($\mathbf{A} > 0$) that is mitigated by negative equilibrium spillovers ($\mathbf{B} < 0$ and $\mathbf{C} < 0$). Our decomposition shows that absent equilibrium effects (i.e., for $\mathbf{B} = \mathbf{C} = 0$), the impact of the reform on the worker match rate would be 1.3 percentage points. Absent equilibrium effects, the impact of the policy reform would therefore be 44% larger, $\frac{(p'_{vr} - p_{vr})\mathbf{A}}{m_W(p'_{vr}) - m_W(p_{vr})} = 1.44$.

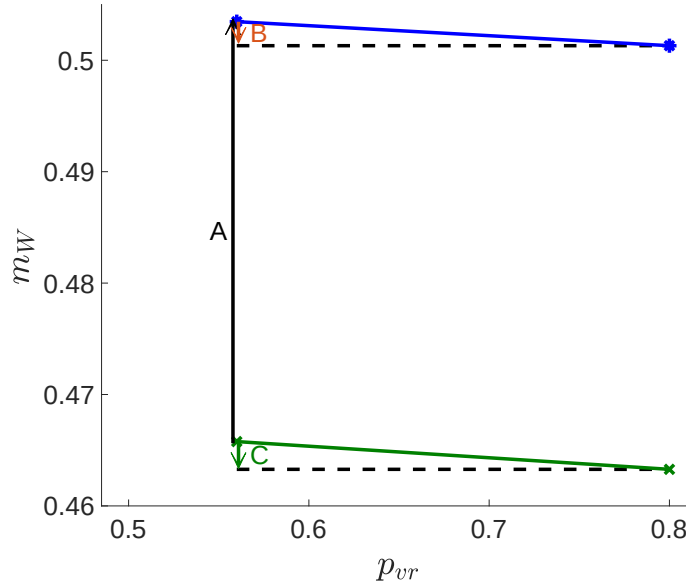
An alternative way to assess the magnitude of equilibrium effects is to compare the individual-level effect of assigning a VR to a single individual with the effects of the considered reform on newly treated, previously treated, and untreated individuals. Based on our estimated model, the treatment effect of assigning a VR to a single individual, $\mathbf{A}$, equals 3.8 percentage points. In contrast, the average effect of the reform on newly treated individuals, $\mathbf{A} + \mathbf{B}$, is about 10 %

¹⁵In the language of the treatment evaluation literature (see, e.g., Imbens and Angrist (1994) and Heckman and Vytlačil (2007)), the first component corresponds to a treatment effect defined under the assumption that equilibrium spillovers are absent (the standard Stable Unit Treatment Value Assumption, SUTVA). The second and third components capture equilibrium effects on treated and untreated individuals, respectively, which lie outside the scope of standard treatment evaluation frameworks.

smaller, at 3.4 percentage points. In addition, the reform reduces the match rates of individuals who were already treated prior to the reform by 0.35 percentage points (**B**) and of untreated individuals by 0.4 percentage points (**C**).

These magnitude suggests that estimates from small-scale quasi-experimental designs that neglect equilibrium effects may substantially overstate the aggregate impact of vacancy referrals.

Figure 4: Worker match rates, given VR/ no VR



7 Conclusion

In this paper, we analyze the equilibrium effects of VRs. Our results show that assigning VRs to workers imposes negative equilibrium spillovers on other workers by lowering their match probabilities. These equilibrium effects substantially attenuate the impact of VRs on aggregate unemployment. As a result, small-scale quasi-experiments that identify individual-level effects of VRs on reemployment probabilities may substantially overstate the effect of VRs on the economywide employment rate.

An interesting extension would be to study the role of on-the-job search, which in some segments of the labor market may generate additional search activity and thereby larger applicant pools competing for vacancies. Another natural direction for future research, would be allowing for recruitment through channels other than officially posted vacancies.

References

- Abbring, Jaap and James Heckman (2007). “Econometric Evaluation of Social Programs, Part III: Distributional Treatment Effects, Dynamic Treatment Effects, Dynamic Discrete Choice, and General Equilibrium Policy Evaluation”. In: *Handbook of Econometrics*. Ed. by J.J. Heckman and E.E. Leamer. 1st ed. Vol. 6B. Elsevier. Chap. 72.
- Albrecht, James, Pieter A. Gautier, and Susan Vroman (2006). “Equilibrium Directed Search with Multiple Applications”. In: *The Review of Economic Studies* 73.4, pp. 869–891. ISSN: 00346527, 1467937X.
- Albrecht, James, Serene Tan, Pieter Gautier, and Susan Vroman (2004). “Matching with multiple applications revisited”. In: *Economics Letters* 84.3, pp. 311–314. ISSN: 0165-1765. DOI: <https://doi.org/10.1016/j.econlet.2003.12.024>.
- Arni, Patrick, Marco Caliendo, Steffen Künn, and Klaus F. Zimmermann (2014). “The IZA evaluation dataset survey: a scientific use file”. In: *IZA Journal of European Labor Studies* 3.1, p. 6. DOI: 10.1186/2193-9012-3-6.
- Boeri, Tito and Herbert Bruecker (Aug. 2014). “Short-time work benefits revisited: some lessons from the Great Recession”. In: *Economic Policy* 26.68, pp. 697–765. ISSN: 0266-4658. DOI: 10.1111/j.1468-0327.2011.271.x. eprint: <https://academic.oup.com/economicpolicy/article-pdf/26/68/697/9571060/economicpolicy26-0697.pdf>.
- Bollens, Joost and Bart Cockx (2017). “Effectiveness of a job vacancy referral scheme”. In: *IZA Journal of Labor Policy* 6.1, p. 15. ISSN: 2193-9004. DOI: 10.1186/s40173-017-0094-0.
- Burda, Michael C. and Jennifer Hunt (2011). “What Explains the German Labor Market Miracle in the Great Recession”. In: *Brookings Papers on Economic Activity* 42.1 (Spring, pp. 273–335. DOI: None.
- Caliendo, Marco, Armin Falk, Lutz C. Kaiser, Hilmar Schneider, Arne Uhlenhorff, Gerard van den Berg, and Klaus F. Zimmermann (2011). “The IZA Evaluation Dataset: towards evidence-based labor policy making”. In: *International Journal of Manpower* 32.7, pp. 731–752. DOI: 10.1108/01437721111174730. eprint: <https://doi.org/10.1108/01437721111174730>.
- Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora (Apr. 2013). “Do Labor Market Policies have Displacement Effects? Evidence from a Clustered Randomized Experiment *”. In: *The Quarterly Journal of Economics* 128.2, pp. 531–580.
- Engström, Per, Patrik Hesselius, and Bertil Holmlund (2012). “Vacancy Referrals, Job Search, and the Duration of Unemployment: A Randomized Experiment”. In: *LABOUR* 26.4, pp. 419–435.
- Fougère, Denis, Jacqueline Pradel, and Muriel Roger (2009). “Does the public employment service affect search effort and outcomes?” In: *European Economic Review* 53.7, pp. 846–869.
- Gautier, Pieter A., José L. Moraga-González, and Ronald P. Wolthoff (2016). “Search costs and efficiency: Do unemployed workers search enough?” In: *European Economic Review* 84. European Labor Market Issues, pp. 123–139. ISSN: 0014-2921. DOI: <https://doi.org/10.1016/j.euroecorev.2015.04.001>.
- Gautier, Pieter, Paul Muller, Bas van der Klaauw, Michael Rosholm, and Michael Svarer (2018). “Estimating Equilibrium Effects of Job Search Assistance”. In: *Journal of Labor Economics* 36.4, pp. 1073–1125.

- Grubb, David (Jan. 2001). “Eligibility Criteria for Unemployment Benefits”. In: *OECD Economic Studies (Paris)* 31.
- Heckman, James J. and Edward J. Vytlacil (2007). “Econometric Evaluation of Social Programs, Part I: Causal Models, Structural Models and Econometric Policy Evaluation”. In: *Handbook of Econometrics*. Ed. by James J. Heckman and Edward E. Leamer. Vol. 6, Part 2. Elsevier. Chap. 70, pp. 4779–4874.
- Imbens, Guido W. and Joshua D. Angrist (1994). “Identification and Estimation of Local Average Treatment Effects”. In: *Econometrica* 62.2, pp. 467–475.
- Immervoll, Herwig and Carlo Knotz (2018). “How demanding are activation requirements for jobseekers”. In: *OECD Social, Employment and Migration Working Papers* 215.
- Kubis Alexander; Moczall, Andreas; Rebien Martina (2018). “The IAB Job Vacancy Survey. Establishment survey on labour demand and recruitment processes Waves 2000 to 2015 and subsequent quarters since 2006”. In: *FDZ-Datenreport, 05/2018 (en)*, Nuremberg.
- Lehmann, Tobias, Camille Terrier, and Rafael Lalive (2023). “Costs and Benefits of Congestion in Two-Sided Markets: Evidence from the Dating Market”. In: 964. DOI: None.
- Möller, Joachim (2010). “The German labor market response in the world recession : de-mystifying a miracle”. In: *Zeitschrift für ArbeitsmarktForschung - Journal for Labour Market Research* 42.4, pp. 325–336. DOI: 10.1007/s12651-009-0026-6.
- Petrongolo, Barbara and Christopher A. Pissarides (2001). “Looking into the Black Box: A Survey of the Matching Function”. In: *Journal of Economic Literature* 39.2, pp. 390–431. DOI: 10.1257/je1.39.2.390.
- Rogerson, Richard, Robert Shimer, and Randall Wright (2005). “Search-Theoretic Models of the Labor Market: A Survey”. In: *Journal of Economic Literature* 43.4, pp. 959–988. DOI: 10.1257/002205105775362014.
- van den Berg, Gerard J., Hanno Foerster, and Arne Uhlenborff (2019). “A Structural Analysis of Vacancy Referrals with Imperfect Monitoring and Sickness”. In: *working paper*.
- van den Berg, Gerard J., Barbara Hofmann, and Arne Uhlenborff (2014). “The Role of Sickness in the Evaluation of Job Search Assistance and Sanctions”.